

Profit Maximization, supply and Market Equilibrium

ECON 101: Introduction to
Microeconomics

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Perfectly Competitive Markets

PRICE TAKING

Because each individual firm sells a sufficiently small proportion of total market output, its decisions have no impact on market price.

price taker Firm that has no influence over market price and thus takes the price as given.

PRODUCT HOMOGENEITY

When the products of all of the firms in a market are perfectly substitutable with one another—that is, when they are homogeneous—no firm can raise the price of its product above the price of other firms without losing most or all of its business.

In contrast, when products are heterogeneous, each firm has the opportunity to raise its price above that of its competitors without losing all of its sales.

The assumption of product homogeneity is important because it ensures that there is a single market price, consistent with supply-demand analysis.

Perfectly Competitive Markets

FREE ENTRY AND EXIT

free entry (or exit) Condition under which there are no special costs that make it difficult for a firm to enter (or exit) an industry.

With free entry and exit, *buyers can easily switch from one supplier to another, and suppliers can easily enter or exit a market.*

When Is a Market Highly Competitive?

Many markets are *highly* competitive in the sense that firms face highly elastic demand curves and relatively easy entry and exit, but there is no simple rule of thumb to describe whether a market is close to being perfectly competitive.

Because firms can implicitly or explicitly collude in setting prices, the presence of many firms is not sufficient for an industry to approximate perfect competition.

Conversely, the presence of only a few firms in a market does not rule out competitive behavior.

Profit Maximization

Do Firms Maximize Profit?

The assumption of *profit maximization* predicts business behavior reasonably accurately and avoids unnecessary analytical complications.

For smaller firms, profit is likely to dominate almost all decisions. In larger firms, however, managers who make day-to-day decisions usually have little contact with the owners.

Managers may be more concerned with such goals as revenue maximization, revenue growth

Firms that do not come close to maximizing profit are not likely to survive. The firms that do survive make long-run profit maximization one of their highest priorities.

Marginal Revenue, Marginal Cost, and Profit Maximization

profit Difference between total revenue and total cost.

$$\Pi(q) = R(q) - C(q)$$

marginal revenue Change in revenue resulting from a one-unit increase in output.

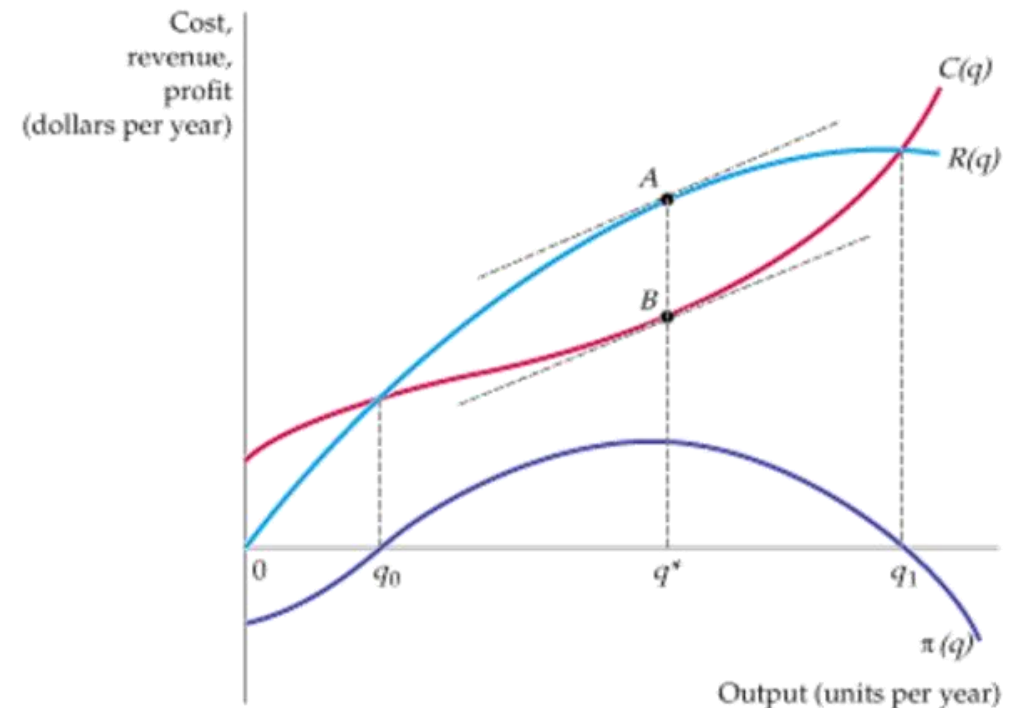
PROFIT MAXIMIZATION IN THE SHORT RUN

A firm chooses output q^* , so that profit, the difference AB between revenue R and cost C , is maximized.

At that output, marginal revenue (the slope of the revenue curve) is equal to marginal cost (the slope of the cost curve).

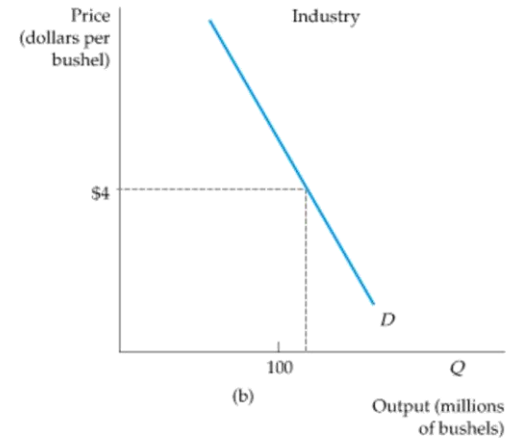
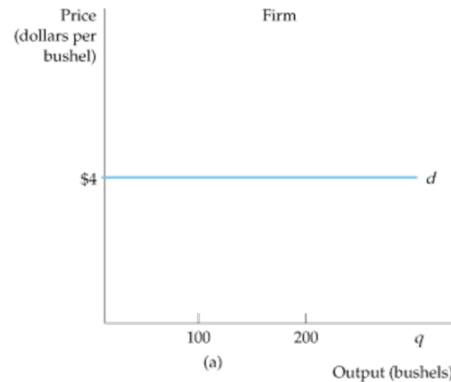
$$\Delta\Pi/\Delta q = \Delta R/\Delta q - \Delta C/\Delta q = 0$$

$$MR(q) = MC(q)$$



Marginal Revenue, Marginal Cost, and Profit Maximization

Demand and Marginal Revenue for a Competitive Firm



DEMAND CURVE FACED BY A COMPETITIVE FIRM

A competitive firm supplies only a small portion of the total output of all the firms in an industry. Therefore, the firm takes the market price of the product as given, choosing its output on the assumption that the price will be unaffected by the output choice. In **(a)** the demand curve facing the firm is perfectly elastic, even though the market demand curve in **(b)** is downward sloping.

Marginal Revenue, Marginal Cost, and Profit Maximization

Because each firm in a competitive industry sells only a small fraction of the entire industry output, *how much output the firm decides to sell will have no effect on the market price of the product.*

Because it is a price taker, *the demand curve facing an individual competitive firm is given by a horizontal line.*

The demand d curve facing an individual firm in a competitive market is both its average revenue and its marginal revenue curve. Along this demand curve, marginal revenue, average revenue, and price are all equal.

Short-Run Profit Maximization by a Competitive Firm

A perfectly competitive firm should choose its output so that *marginal cost equals price:*

$$MC(q) = MR = P$$

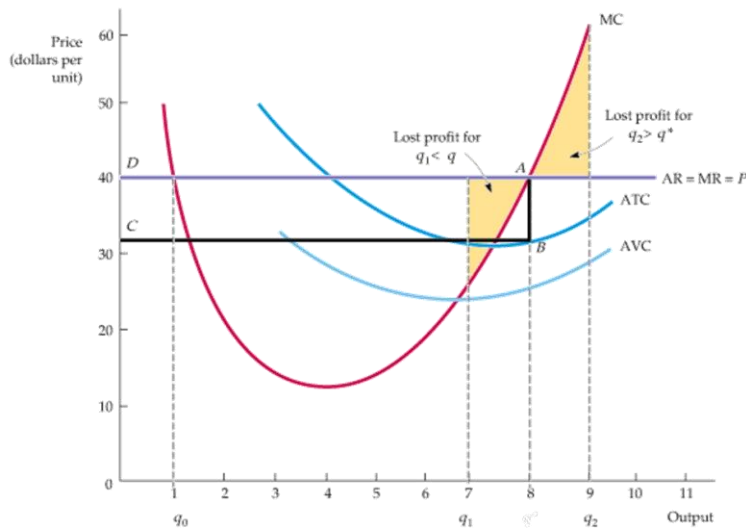
Choosing Output in the Short Run

A COMPETITIVE FIRM MAKING A POSITIVE PROFIT

In the short run, the competitive firm maximizes its profit by choosing an output q^* at which its marginal cost MC is equal to the price P (or marginal revenue MR) of its product.

The profit of the firm is measured by the rectangle $ABCD$.

Any change in output, whether lower at q_1 or higher at q_2 , will lead to lower profit.



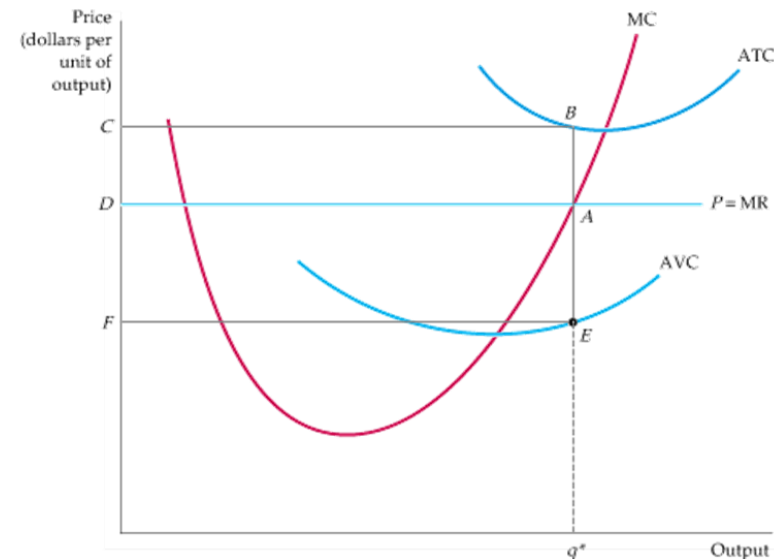
Output Rule: If a firm is producing any output, it should produce at the level at which marginal revenue equals marginal cost.

When Should the Firm Shut Down?

A COMPETITIVE FIRM INCURRING LOSSES

A competitive firm should shut down if price is below AVC .

The firm may produce in the short run if price is greater than average variable cost.



THE SHORT-RUN OUTPUT DECISION OF AN ALUMINUM SMELTING PLANT

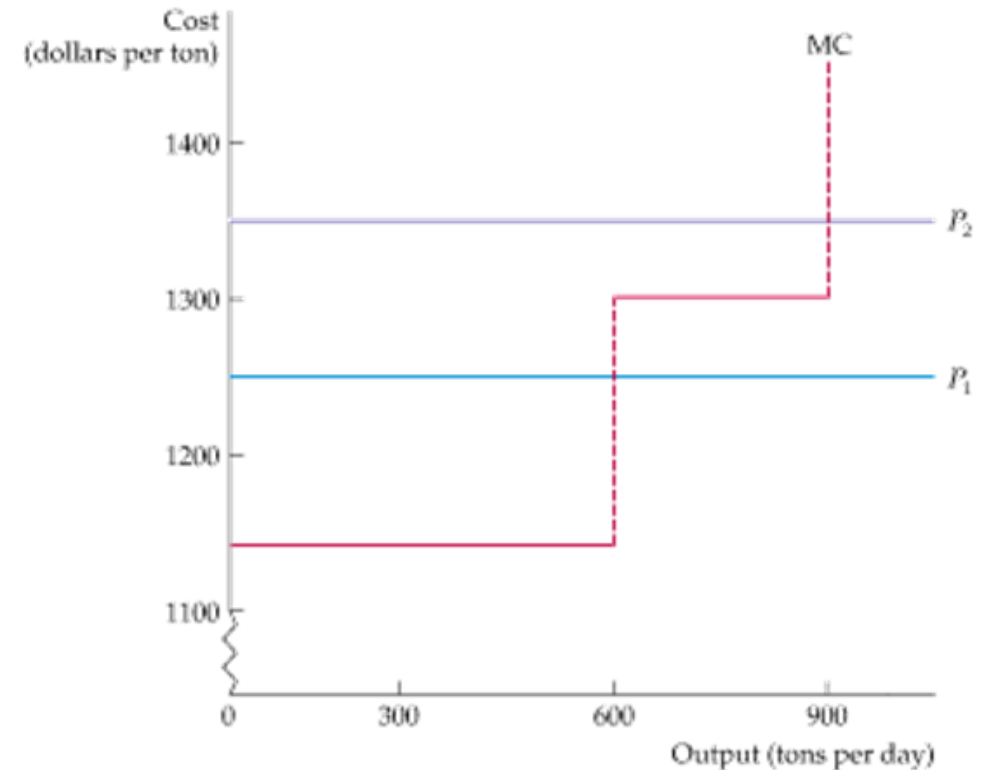
How should the manager determine the plant's profit maximizing output? Recall that the smelting plant's short-run marginal cost of production depends on whether it is running two or three shifts per day.

THE SHORT-RUN OUTPUT OF AN ALUMINUM SMELTING PLANT

In the short run, the plant should produce 600 tons per day if price is above \$1140 per ton but less than \$1300 per ton.

If price is greater than \$1300 per ton, it should run an overtime shift and produce 900 tons per day.

If price drops below \$1140 per ton, the firm should stop producing, but it should probably stay in business because the price may rise in the future.



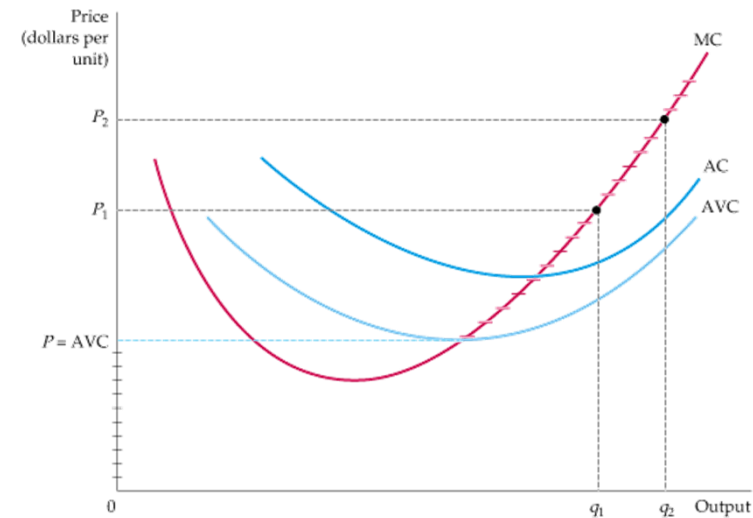
The Competitive Firm's Short-run Supply Curve

The firm's supply curve is the portion of the marginal cost curve for which marginal cost is greater than average variable cost.

THE SHORT-RUN SUPPLY CURVE FOR A COMPETITIVE FIRM

In the short run, the firm chooses its output so that marginal cost MC is equal to price as long as the firm covers its average variable cost.

The short-run supply curve is given by the crosshatched portion of the marginal cost curve.

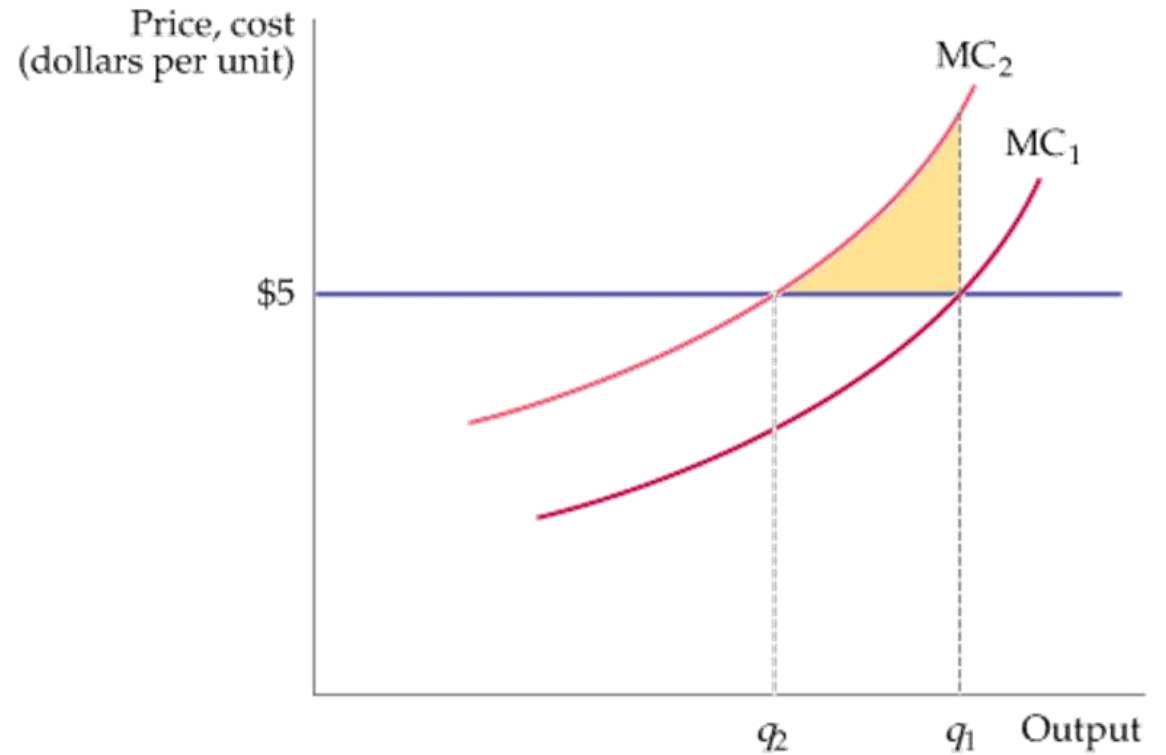


The Competitive Firm's Short-run Supply Curve

The Firm's Response to an Input Price Change

THE RESPONSE OF A FIRM TO A CHANGE IN INPUT PRICE

When the marginal cost of production for a firm increases (from MC_1 to MC_2), the level of output that maximizes profit falls (from q_1 to q_2).



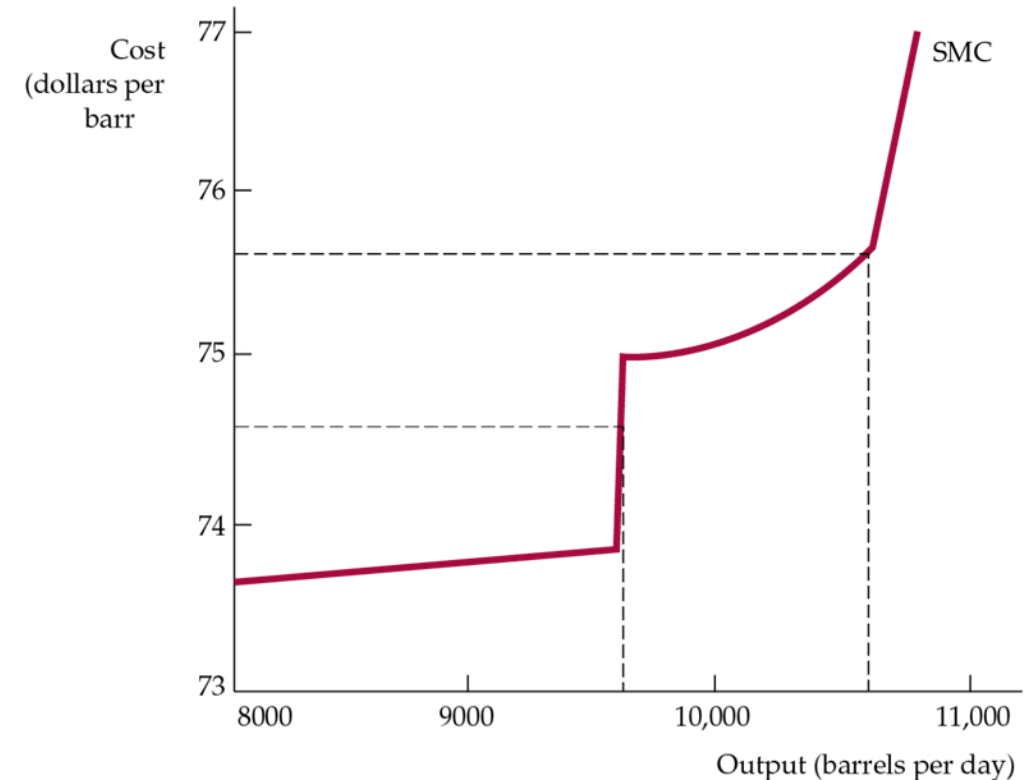
THE SHORT-RUN PRODUCTION OF PETROLEUM PRODUCTS

Although plenty of crude oil is available, the amount that you refine depends on the capacity of the refinery and the cost of production.

THE SHORT-RUN PRODUCTION OF PETROLEUM PRODUCTS

As the refinery shifts from one processing unit to another, the marginal cost of producing petroleum products from crude oil increases sharply at several levels of output.

As a result, the output level can be insensitive to some changes in price but very sensitive to others.



The Short-Run Market Supply Curve

INDUSTRY SUPPLY IN THE SHORT RUN

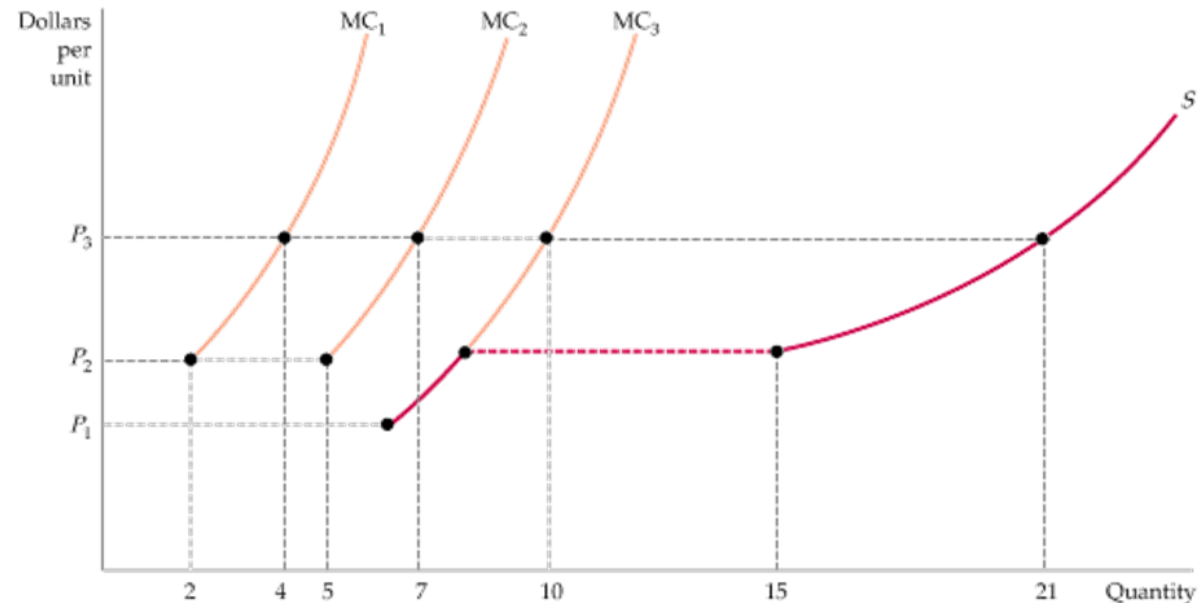
The short-run industry supply curve is the summation of the supply curves of the individual firms.

Because the third firm has a lower average variable cost curve than the first two firms, the market supply curve S begins at price P_1 and follows the marginal cost curve of the third firm MC_3 until price equals P_2 , when there is a kink.

For P_2 and all prices above it, the industry quantity supplied is the sum of the quantities supplied by each of the three firms.

Elasticity of Market Supply

$$E_s = (\Delta Q/Q)/(\Delta P/P)$$



THE SHORT-RUN WORLD SUPPLY OF COPPER

Costs of mining, smelting, and refining copper differ because of differences in labor and transportation costs and because of differences in the copper content of the ore.

THE WORLD COPPER INDUSTRY (2014)

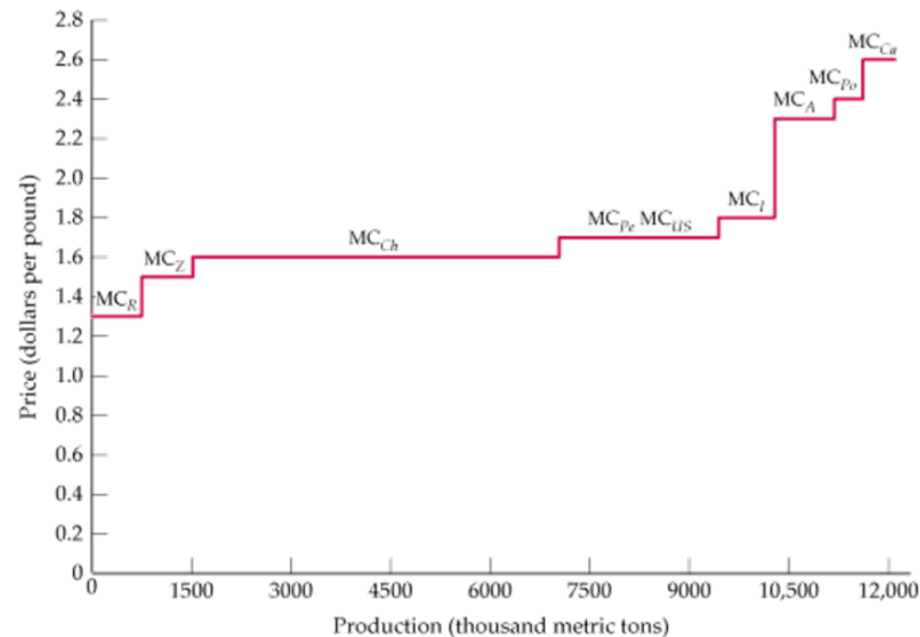
COUNTRY	ANNUAL PRODUCTION (THOUSAND METRIC TONS)	MARGINAL COST (DOLLARS PER POUND)
Australia	1000	2.30
Canada	680	2.60
Chile	5800	1.60
Indonesia	400	1.80
Peru	1400	1.70
Poland	425	2.40
Russia	850	1.30
US	1370	1.70
Zambia	730	1.50

THE SHORT-RUN WORLD SUPPLY OF COPPER

The world supply curve is obtained by summing each nation's supply curve horizontally. The elasticity of supply depends on the price of copper. At relatively low prices, the curve is quite elastic because small price increases lead to large increases in the quantity of copper supplied. At higher prices—say, above \$2.40 per pound—the curve becomes more inelastic because, at those prices, most producers would be operating close to or at capacity.

The supply curve for world copper is obtained by summing the marginal cost curves for each of the major copper-producing countries.

The supply curve slopes upward because the marginal cost of production ranges from a low of 1.30 in Russia to a high of \$2.60 in Canada.



The Short-Run Market Supply Curve

PRODUCER SURPLUS VERSUS PROFIT

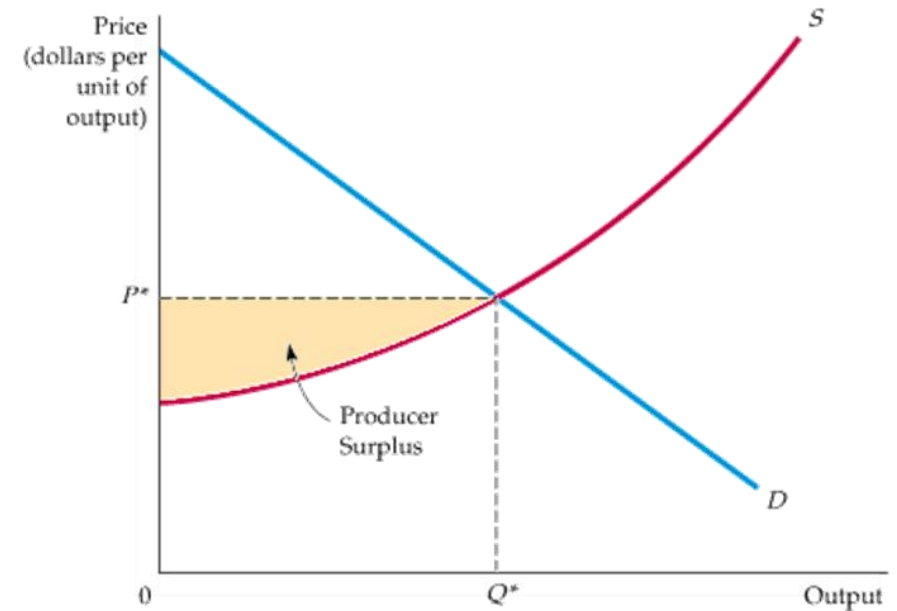
$$\text{Producer surplus} = PS = R - VC$$

$$\text{Profit} = \Pi = R - VC - FC$$

FIGURE 8.12

PRODUCER SURPLUS FOR A MARKET

The producer surplus for a market is the area below the market price and above the market supply curve, between 0 and output Q^* .



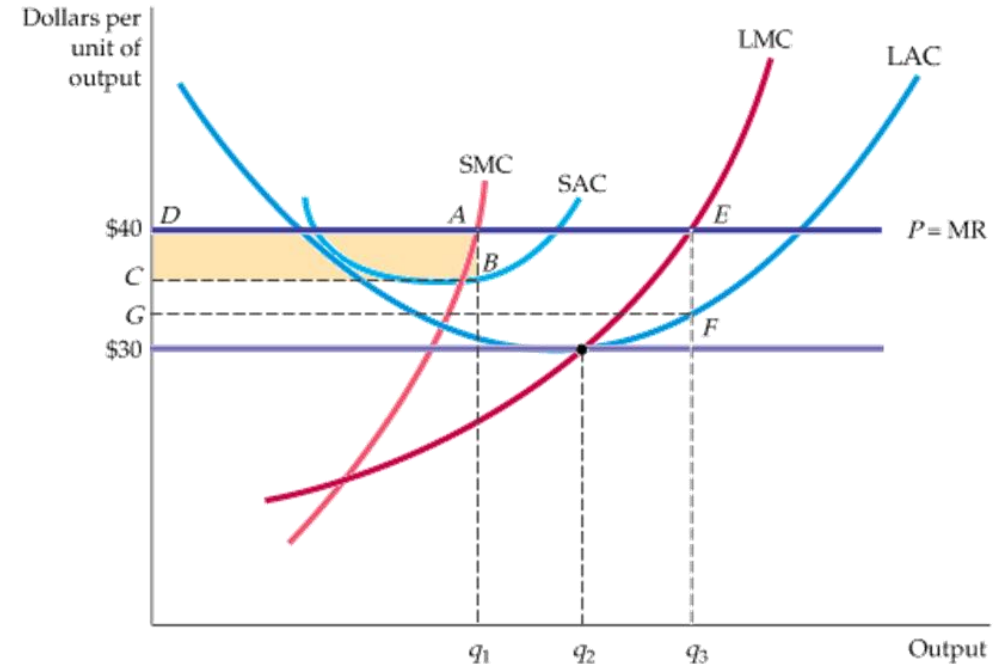
Choosing Output in the Long Run

Long-Run Profit Maximization

OUTPUT CHOICE IN THE LONG RUN

The firm maximizes its profit by choosing the output at which price equals long-run marginal cost LMC.

In the diagram, the firm increases its profit from $ABCD$ to $EFGD$ by increasing its output in the long run.



The long-run output of a profit-maximizing competitive firm is the point at which long-run marginal cost equals the price.

Choosing Output in the Long Run

Long-Run Competitive Equilibrium

ACCOUNTING PROFIT AND ECONOMIC PROFIT

Economic profit takes into account opportunity costs. One such opportunity cost is the return to the firm's owners if their capital were used elsewhere. Accounting profit equals revenues R minus labor cost wL , which is positive. Economic profit π , however, equals revenues R minus labor cost wL minus the capital cost, rK .

$$\Pi = R - wL - rK$$

ZERO ECONOMIC PROFIT

zero economic profit A firm is earning a normal return on its investment—i.e., it is doing as well as it could by investing its money elsewhere.

ENTRY AND EXIT

In a market with entry and exit, a firm enters when it can earn a positive long-run profit and exits when it faces the prospect of a long-run loss.

Choosing Output in the Long Run

long-run competitive equilibrium All firms in an industry are maximizing profit, no firm has an incentive to enter or exit, and price is such that quantity supplied equals quantity demanded.

When a firm earns zero economic profit, it has no incentive to exit the industry. Likewise, other firms have no special incentive to enter.

A **long-run competitive equilibrium** occurs when three conditions hold:

1. All firms in the industry are maximizing profit.
2. No firm has an incentive either to enter or exit the industry because all firms are earning zero economic profit.
3. The price of the product is such that the quantity supplied by the industry is equal to the quantity demanded by consumers.

Choosing Output in the Long Run

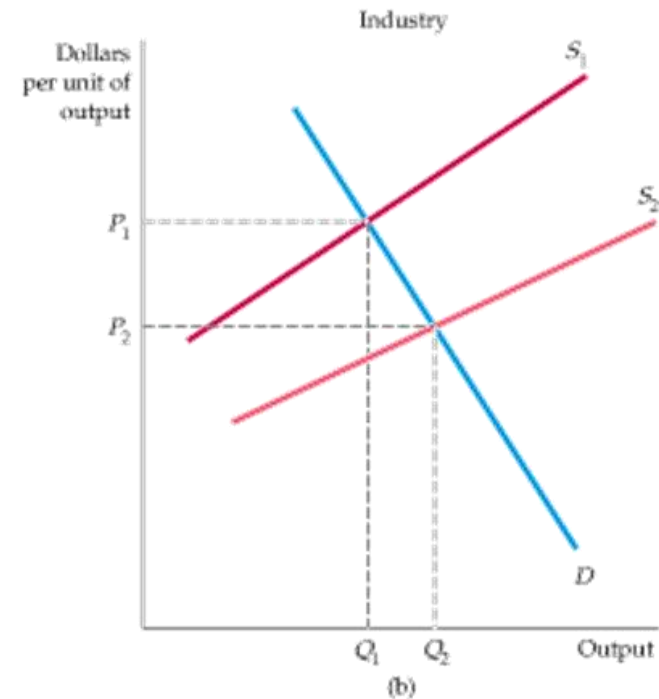
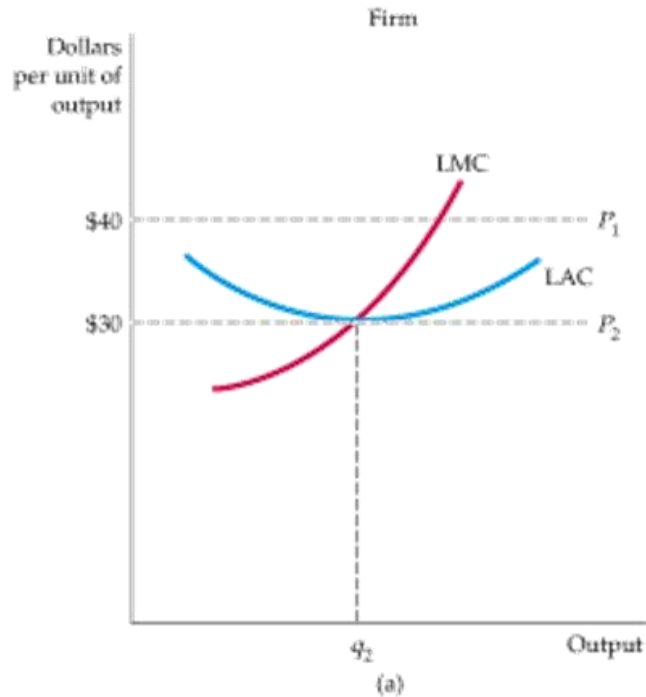
LONG-RUN COMPETITIVE EQUILIBRIUM

Initially the long-run equilibrium price of a product is \$40 per unit, shown in **(b)** as the intersection of demand curve D and supply curve S_1 .

In **(a)** we see that firms earn positive profits because long-run average cost reaches a minimum of \$30 (at q_2).

Positive profit encourages entry of new firms and causes a shift to the right in the supply curve to S_2 , as shown in **(b)**.

The long-run equilibrium occurs at a price of \$30, as shown in **(a)**, where each firm earns zero profit and there is no incentive to enter or exit the industry.



Choosing Output in the Long Run

FIRMS HAVING IDENTICAL COSTS

To see why all the conditions for long-run equilibrium must hold, assume that all firms have identical costs.

If too many firms enter the industry in response to an opportunity for profit, the industry supply curve will shift to the right, and price will fall.

Only when there is no incentive to exit or enter can a market be in long-run equilibrium.

FIRMS HAVING DIFFERENT COSTS

Perhaps one firm has a patent that lets it produce at a lower average cost than all the others. In that case, it is consistent with long-run equilibrium for that firm to earn a greater *accounting* profit and to enjoy a higher producer surplus than other firms.

If the patent is profitable, other firms in the industry will pay to use it. The increased value of the patent thus represents an opportunity cost to the firm that holds it. It could sell the rights to the patent rather than use it. If all firms are equally efficient otherwise, the *economic* profit of the firm falls to zero.

Choosing Output in the Long Run

THE OPPORTUNITY COST OF LAND

There are instances in which firms earning positive accounting profit may be earning zero economic profit.

If a clothing store is located near a large shopping center, the additional flow of customers can increase the store's accounting profit. When the opportunity cost of the land is included, the profitability of the clothing store is no higher than that of its competitors.

Economic Rent

economic rent Amount that firms are willing to pay for an input less the minimum amount necessary to obtain it.

In competitive markets, in both the short and the long run, economic rent is often positive even though profit is zero.

Producer Surplus in the Long Run

In the long run, in a competitive market, *the producer surplus that a firm earns on the output that it sells consists of the economic rent that it enjoys from all its scarce inputs.*

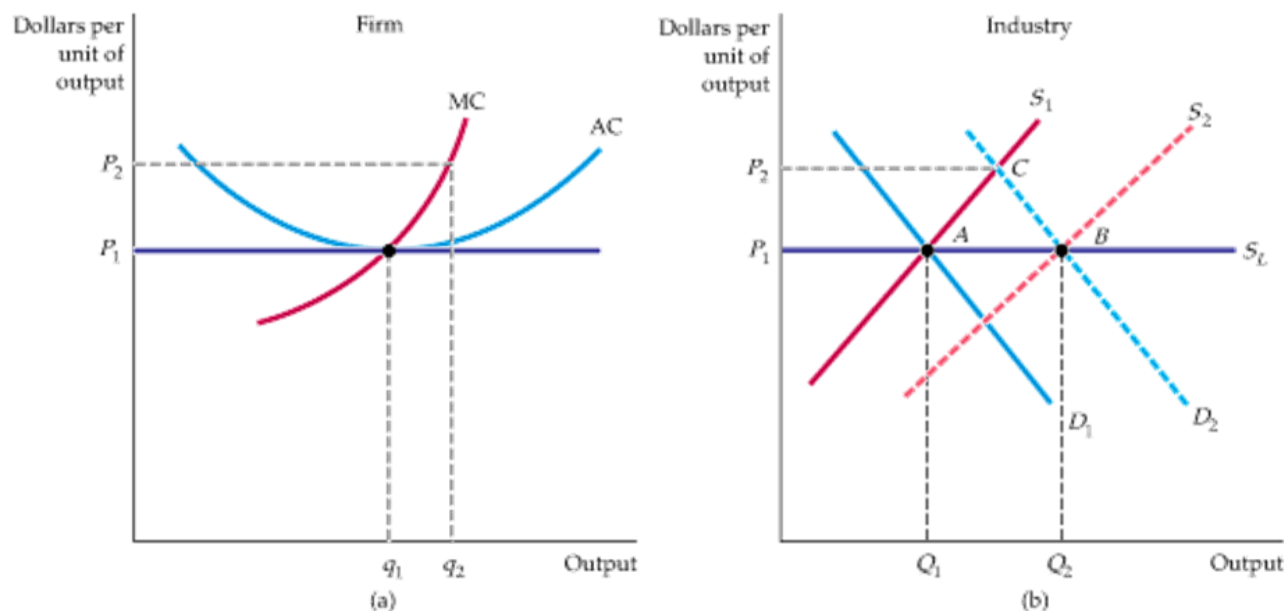
The Industry's Long-Run Supply Curve

Constant-Cost Industry

constant-cost industry Industry whose long-run supply curve is horizontal.

LONG-RUN SUPPLY IN A CONSTANT COST INDUSTRY

In **(b)**, the long-run supply curve in a constant-cost industry is a horizontal line S_L . When demand increases, initially causing a price rise, the firm initially increases its output from q_1 to q_2 , as shown in **(a)**. But the entry of new firms causes a shift to the right in industry supply. Because input prices are unaffected by the increased output of the industry, entry occurs until the original price is obtained (at point B in **(b)**).



The long-run supply curve for a constant-cost industry is, therefore, a horizontal line at a price that is equal to the long-run minimum average cost of production.

The Industry's Long-Run Supply Curve

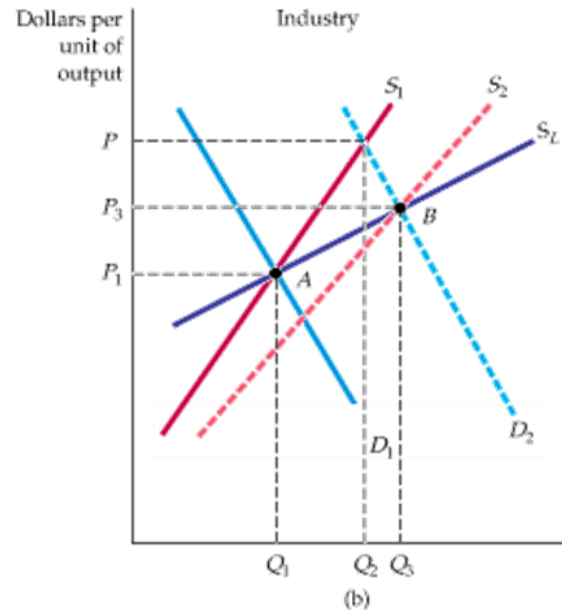
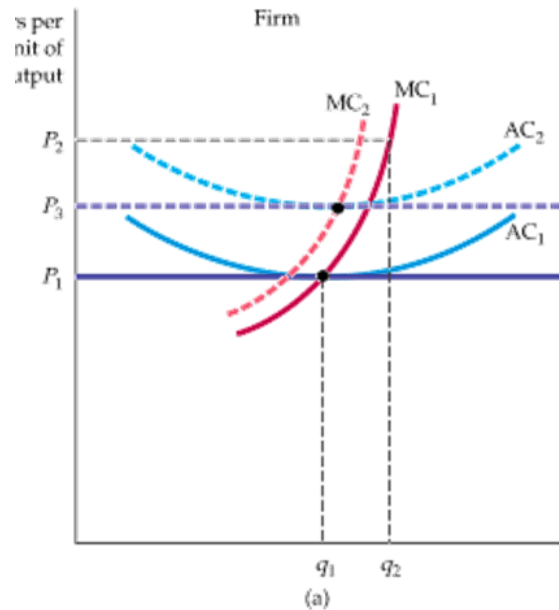
Increasing-Cost Industry

increasing-cost industry Industry whose long-run supply curve is upward sloping.

LONG-RUN SUPPLY IN A CONSTANT COST INDUSTRY

In **(b)**, the long-run supply curve in an increasing-cost industry is an upward-sloping curve S_L . When demand increases, initially causing a price rise, the firms increase their output from q_1 to q_2 in **(a)**. In that case, the entry of new firms causes a shift to the right in supply from S_1 to S_2 .

Because input prices increase as a result, the new long-run equilibrium occurs at a higher price than the initial equilibrium.



In an increasing-cost industry, the long-run industry supply curve is upward sloping.

The Industry's Long-Run Supply Curve

Decreasing-Cost Industry

decreasing-cost industry Industry whose long-run supply curve is downward sloping.

CONSTANT-, INCREASING-, AND DECREASING-COST INDUSTRIES: COFFEE, OIL, AND AUTOMOBILES

You have been introduced to industries that have constant, increasing, and decreasing long-run costs.

We saw that the supply of coffee is extremely elastic in the long run. The reason is that land for growing coffee is widely available and the costs of planting and caring for trees remains constant as the volume grows. Thus, coffee is a constant-cost industry.

The oil industry is an increasing cost industry because there is a limited availability of easily accessible, large-volume oil fields.

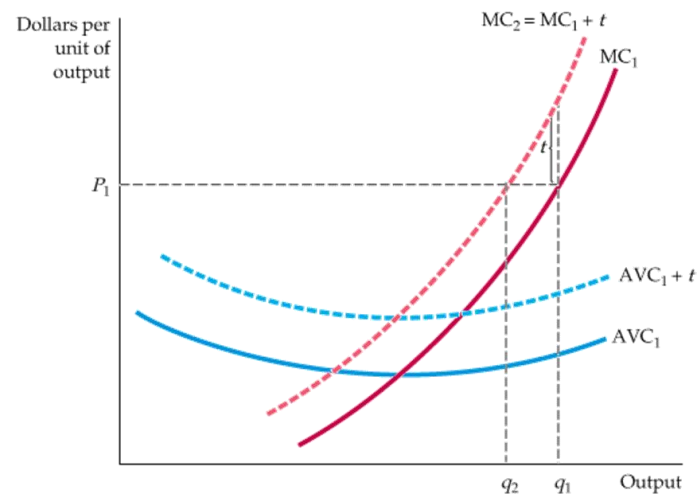
Finally, a decreasing-cost industry. In the automobile industry, certain cost advantages arise because inputs can be acquired more cheaply as the volume of production increases.

The Effects of a Tax

EFFECT OF AN OUTPUT TAX ON A COMPETITIVE FIRM'S OUTPUT

An output tax raises the firm's marginal cost curve by the amount of the tax.

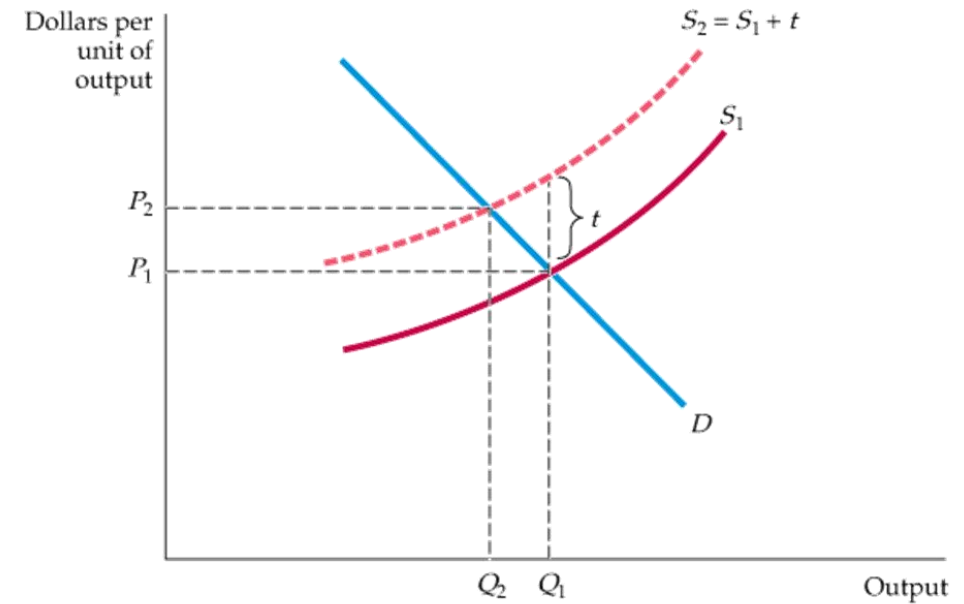
The firm will reduce its output to the point at which the marginal cost plus the tax is equal to the price of the product.



EFFECT OF AN OUTPUT TAX ON INDUSTRY OUTPUT

An output tax placed on all firms in a competitive market shifts the supply curve for the industry upward by the amount of the tax.

This shift raises the market price of the product and lowers the total output of the industry.



The Industry's Long-Run Supply Curve

Long-Run Elasticity of Supply

The long-run elasticity of industry supply is defined in the same way as the short-run elasticity: It is the percentage change in output ($\Delta Q/Q$) that results from a percentage change in price ($\Delta P/P$).

In a constant-cost industry, the long-run supply curve is horizontal, and the long-run supply elasticity is infinitely large. (A small increase in price will induce an extremely large increase in output.) In an increasing-cost industry, however, the long-run supply elasticity will be positive but finite.

Because industries can adjust and expand in the long run, we would generally expect long-run elasticities of supply to be larger than short-run elasticities.

The magnitude of the elasticity will depend on the extent to which input costs increase as the market expands. For example, an industry that depends on inputs that are widely available will have a more elastic long-run supply than will an industry that uses inputs in short supply.

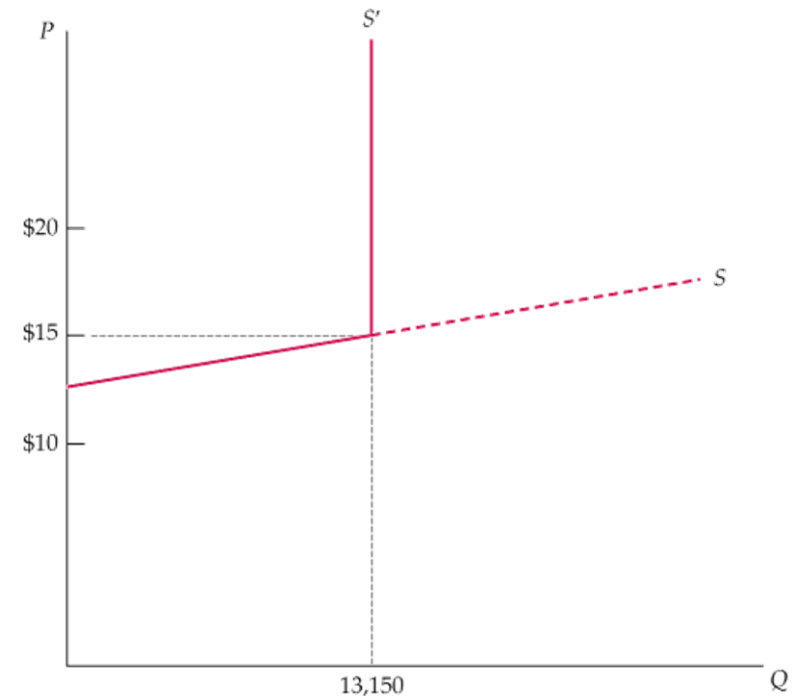
THE SUPPLY OF TAXICABS IN NEW YORK

While reducing taxi fares will indeed cause a reduction in the quantity supplied, raising the price will not cause an increase in the quantity supplied. Why not? Because the number of medallions is fixed.

THE SUPPLY CURVE FOR NEW YORK TAXICABS

If there were no restriction on the number of medallions, the supply curve would be highly elastic. Cab drivers work hard and don't earn much, so a drop in the price P (of a 5-mile ride) would lead many of them to find another job. Likewise, an increase in price would bring many new drivers into the market.

But the number of medallions—and therefore the number of taxicabs—is limited to 13,150, so the supply curve becomes vertical at this quantity.



THE LONG-RUN SUPPLY OF HOUSING

To begin, consider the supply of owner-occupied housing in suburban or rural areas where land is not scarce. In this case, the price of land does not increase substantially as the quantity of housing supplied increases.

Likewise, costs associated with construction are not likely to increase because there is a national market for lumber and other materials. Therefore, the long-run elasticity of the housing supply is likely to be very large, approximating that of a constant-cost industry.

The market for rental housing is different, however. The construction of rental housing is often restricted by local zoning laws. Many communities outlaw it entirely, while others limit it to certain areas. Because urban land on which most rental housing is located is restricted and valuable, the long-run elasticity of supply of rental housing is much lower than the elasticity of supply of owner-occupied housing. With urban land becoming more valuable as housing density increases, and with the cost of construction soaring, increased demand causes the input costs of rental housing to rise. In this increasing-cost case, the elasticity of supply can be much less than 1.



End of this Lecture.

Based on Chapters 8 of Pindyck and Rubinfeld.